

Monitoring Models Before the Labels Arrive: A Benchmark for Performance Estimation, Drift Detection and Retraining under Natural Drift and Label Delay

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Abstract

Deployed models are monitored long before their true labels arrive: a mortgage default is known up to two years after the loan is made. Yet drift detectors and label-free performance estimators are usually evaluated on artificially injected drift with labels available immediately. We present an open benchmark that replays five real, timestamped data sources (Freddie Mac mortgages, US Census ACS, CDC BRFSS, NYC taxi trips and TabReD) as streams in which a monitor sees each label only when it would have arrived in practice, including the *natural* label delay of mortgage defaults. The benchmark scores four tasks on eight experiments: estimating the performance of a new batch, estimating the performance of the partly labeled “in-flight” book, detecting degradation, and deciding when to retrain, the last scored as regret against a hindsight-optimal schedule. Three findings stand out. (1) Input-drift tests (PSI, Kolmogorov–Smirnov, MMD, domain classifiers) alarm in every period of the median experiment: on real data the inputs always change. (2) Label-free estimators miss shocks that change outcomes but not inputs (the 2007–08 mortgage crisis), while label-based monitoring sees them about two years late. (3) A fixed retraining calendar beats monitor-triggered retraining in most experiments. We also propose delay-adjusted CBPE, which uses early-arriving labels without being biased by them. It gives the best degradation alarm overall (median rank correlation with true degradation 0.58, precision 0.84). Its chain-ladder form fails, like actuarial chain-ladder methods, under calendar-time shocks such as COVID forbearance; an age–period–cohort hazard variant fixes this and gives the lowest in-flight error on Freddie Mac in three of four cases over five seeds, e.g. 31% lower default-rate error than the best label-based baseline across the 2007–2024 period. Code, data loaders and a leaderboard are released under the MIT license.

1 Introduction

Teams that run models in production ask two questions continually: *is the model still working*, and *should we retrain it*? Answering with labels is often impossible in time. A credit model’s outcome (default within 24 months) is fully known only two years after each decision; insurance conversions, medical outcomes and fraud chargebacks arrive weeks to months late. Monitoring therefore relies on proxies: tests that the input distribution has shifted [9, 17], estimators that predict accuracy from the model’s own confidence [8, 11, 15], and streaming detectors on whatever labels have arrived [1, 7].

These methods are rarely evaluated under the conditions they are meant for. Benchmarks typically inject synthetic drift into shuffled data and reveal each label immediately [3]; distribution-shift benchmarks with natural shift [12, 19] evaluate models, not monitors, and ignore when labels

arrive. Delayed labels have been studied for stream classifiers [10], but not as the central obstacle to monitoring.

We build a benchmark around that obstacle. Our contributions:

1. **Streams with label arrival times.** Five public sources replayed in time order; each row carries the time it is scored and the time its label becomes known. For Freddie Mac mortgages the arrival is natural: a default is known in the month it happens, a repayment only when the 24-month window closes (Section 3).
2. **Four tasks with leakage-free evaluation:** next-batch estimation, in-flight book estimation, degradation detection, and retraining decisions scored against a hindsight-optimal schedule computed by dynamic programming (Section 4).
3. **Delay-adjusted CBPE**, a chain-ladder [14] correction that lets confidence-based estimation use early, biased labels, and an age-period-cohort hazard variant robust to calendar-time shocks (Section 5).
4. **An empirical study** of 11 detectors, 8 estimators and 9 retraining policies on 8 experiments (Section 6), with a public leaderboard.

2 Related work

Drift detection. Two-sample tests on inputs or model outputs [17], kernel tests [9], domain classifiers, and the population stability index used in credit risk flag that *data* changed. Streaming detectors such as ADWIN [1] monitor error rates and therefore need labels; see Gama et al. [7] for a survey.

Label-free performance estimation. ATC [8] and DoC [11] predict accuracy from confidence; CBPE [15] computes expected confusion matrices from calibrated probabilities. All are exact under pure covariate shift with calibration and fail under concept shift; label-shift estimators [13] assume $P(x | y)$ is fixed. None uses the labels that do arrive early.

Benchmarks with natural shift. Folktables [4] exposes year-to-year census shift; TabReD [18] supplies industrial tabular data with time-based splits; WILDS and Wild-Time [12, 19] cover natural shift for model robustness. Our benchmark reuses Folktables and TabReD data but evaluates monitors and adds label arrival.

Delayed outcomes in actuarial science. Insurers estimate claims incurred but not yet reported with chain-ladder development patterns [14]; competing-risk models [5] handle outcomes that can be pre-empted (a loan that repays cannot default). Our method borrows both ideas.

3 Benchmark

Stream format. A stream is a table of prediction events. Row i has features x_i , an event time t_i when the model scores it, an eventual label y_i (possibly never resolved) and a label time $\tau_i \geq t_i$ when y_i becomes known. At any decision time s , a monitor may use y_i only if $\tau_i \leq s$.

Table 1: Experiments. Delay: natural, or simulated constant delay.

Experiment	Target	Rows	Delay	Trained on	Monitored
freddie_2007	default $\leq 24m$	1.36M	natural	1999–2003	2007Q1–2024Q1 (q)
freddie_2016	default $\leq 24m$	1.36M	natural	2010–2012	2016Q1–2024Q1 (q)
acs_income_2016	income $> \$50k$	1.0M	365 d	2014	2016–2024 (y)
brfss_2014	diabetes	1.68M	180 d	2011–mid 2013	2014–2025 (q)
tlc_2019	tip $\geq 25\%$	1.8M	30 d	Feb–May 2019	Jul 2019–Dec 2021 (m)
tabred_homecredit	default	382k	90 d	Jan–May 2019	14 months (m)
tabred_homesite	conversion	261k	30 d	Jan–Aug 2013	20 months (m)
tabred_ecom	repeat purchase	160k	14 d	Mar 1–17 2013	5 weeks (w)

Data. Table 1 lists the streams. Freddie Mac [6]: 1.36M loans originated 1999–2026 (the public 50k-per-year sample); label: 90+ days delinquent or a credit-loss termination within 24 months of first payment; the label time is the month of the event, or the month the loan reaches age 24 or prepays. Age is counted in calendar months: the reported loan age is reset when a loan is modified, which would otherwise let defaults years later count as early ones. ACS [4]: income above \$50k, 2014–2024 (2020 has no standard release), relationship and state codes harmonized across years. BRFSS [2]: diagnosed diabetes, 2011–2024, ordered by interview month, 28 variables harmonized across renamings. NYC taxi [16]: card trips 2019–2021, label: tip $\geq 25\%$ of fare. TabReD [18]: the three binary tasks. Streams other than Freddie Mac receive a simulated constant delay (Table 1).

Training without label leakage. At the training date, a model may only learn from cohorts whose labels have (at least 95%) arrived. Using partly labeled cohorts is a common silent error: early-arriving labels (defaults) are over-represented. The last 12 months of trainable cohorts form the reference window on which monitors are calibrated. The model is a gradient-boosted tree ensemble; it is then frozen.

4 Tasks

T1: next-batch estimation. For each period, estimate the frozen model’s metric (AUC, positive rate, and accuracy where classes are balanced) on the batch scored in that period, using its scores and the labels arrived by the period’s end. Score: mean absolute error (MAE) over periods, with 95% moving-block bootstrap intervals over periods.

T2: in-flight book estimation. For long-horizon labels, a new batch’s outcome depends on events that have not happened yet: Freddie Mac loans made in 2019Q1 defaulted at 4.4% because of a pandemic a year later. The question a lender can answer is the state of its *book*: all loans scored in the last 24 months, part of which are already labeled. Same score as T1.

T3: degradation detection. A period is *degraded* if its true metric is worse than the reference by more than Δ (any change, for positive rate). Each detector returns a drift score and an alarm. Scores: Spearman correlation of the drift score with the true degradation; AUROC for several Δ ; precision and recall of the alarms.

T4: retrain or not. After each period a policy may retrain on the labels arrived so far. We precompute one candidate model per decision point and its loss (log loss) on every later period, so

Algorithm 1 Walk-forward evaluation shared by all tasks

Require: stream $\{(x_i, y_i, t_i, \tau_i)\}$, training date s_0 , period length (month/quarter/year)

- 1: $C \leftarrow$ cohorts (periods) before s_0 whose labels are $\geq 95\%$ arrived by s_0
 - 2: fit model f on C minus its last 12 months; reference window $R \leftarrow$ last 12 months of C
 - 3: calibrate each monitor m on $\{(f(x_i), y_i) : i \in R\}$
 - 4: **for** each period P starting at or after s_0 , in time order **do**
 - 5: $s \leftarrow$ end of P ; $B \leftarrow \{i : t_i \in P\}$; $\mathcal{H} \leftarrow \{i : t_i < \text{start of } P\}$
 - 6: history $\leftarrow \{(f(x_i), t_i)\}_{i \in \mathcal{H}} \cup \{(y_i, \tau_i) : i \in \mathcal{H}, \tau_i \leq s\}$
 - 7: $\hat{v}_m \leftarrow m(\{f(x_i), x_i\}_{i \in B}, \text{history})$ $\triangleright$ estimate, drift score or alarm
 - 8: **if** labels of $\geq 90\%$ of B ever resolve **then**
 - 9: $v \leftarrow$ metric of f on $\{(f(x_i), y_i) : i \in B\}$ $\triangleright$ hidden from every monitor
 - 10: record $(P, m, \hat{v}_m, v)$
-

any policy is evaluated by lookup. Cost = total loss + $\lambda \times$ retrains. We report regret against the hindsight-optimal schedule, found by dynamic programming over which model serves each period, for $\lambda = \rho \cdot g$, where g is the average gain of one retrain under “always retrain” and $\rho \in \{0, 0.25, 1, 4\}$.

Protocol. All four tasks share one walk-forward loop (Algorithm 1). Its only source of information about labels is the set $\mathcal{K}(s) = \{i : \tau_i \leq s\}$, so a monitor cannot see a label before it would have arrived. The true metric of a batch is computed from labels the monitor never sees, and only for batches whose labels eventually resolve (at least 90%); later batches still receive estimates but are not scored. This is the real “labels have not arrived” frontier: for Freddie Mac it begins in 2024Q2.

5 Delay-adjusted CBPE

CBPE treats calibrated scores c_i as the probability each row is positive. Under concept shift the calibration breaks; the labels that would reveal this arrive late, and the first to arrive are biased: defaults arrive within months, repayments only after 24.

Arrival curves. From mature rows (older than the longest observed delay) we estimate $F(a)$, the probability that a positive’s label has arrived by age a , and $G(a)$ for negatives.

Offset. We model the true probability as $p_i = \sigma(\text{logit } c_i + \delta)$ and fit δ by maximum likelihood on the last 12 months of rows. Each row contributes p_i if its label arrived positive, $1 - p_i$ if negative, and

$$1 - p_i F(a_i) - (1 - p_i) G(a_i)$$

if still pending at age a_i . Early defaults therefore move δ at once without being over-counted: a pending row is evidence of a negative only to the extent that a positive would have arrived by now. With constant delay the method reduces to recalibrating on the most recent complete labels.

Estimates. For a new batch, the metric is computed as in CBPE from p_i (expected AUC from all positive–negative pairs; expected accuracy and positive rate in closed form). For the in-flight book, arrived labels count as observed and pending rows use the posterior $P(y_i = 1 \mid \text{pending}) = p_i(1 - F(a_i))/(1 - p_i F(a_i) - (1 - p_i) G(a_i))$. Algorithm 2 summarizes the procedure.

Algorithm 2 Delay-adjusted CBPE at decision time s

Require: calibrator $c(\cdot)$ fit on the reference window; history with arrived labels; batch scores

- 1: $D \leftarrow$ longest observed delay $\tau_i - t_i$ among arrived labels
 - 2: $M \leftarrow$ rows with arrived labels and age $s - t_i \in [D, D + 36 \text{ months})$ $\triangleright$ mature rows
 - 3: $F, G \leftarrow$ empirical CDFs of delays of positives / negatives in M
 - 4: $W \leftarrow$ rows of the last 12 months that are labeled or have $F(\text{age}) > 0.05$
 - 5: $\hat{\delta} \leftarrow \arg \max_{\delta} \sum_{i \in W} \log \ell_i(\delta)$, $p_i(\delta) = \sigma(\text{logit } c(f(x_i)) + \delta)$,
 $\ell_i = p_i$ (arrived positive), $1 - p_i$ (arrived negative), $1 - p_i F(a_i) - (1 - p_i) G(a_i)$ (pending)
 - 6: **new batch:** CBPE with $p_i(\hat{\delta})$: positive rate $\frac{1}{n} \sum p_i$, expected AUC over pairs, expected accuracy
 - 7: **in-flight book:** $q_i = y_i$ if arrived, else $p_i(1 - F(a_i)) / (1 - p_i F(a_i) - (1 - p_i) G(a_i))$; metric from q
-

Expected AUC. Given probabilities q_i that row i is positive and scores f_i ,

$$\mathbb{E}[\text{AUC}] \approx \frac{\sum_{i \neq j} q_i(1 - q_j) (\mathbb{I}[f_i > f_j] + \frac{1}{2} \mathbb{I}[f_i = f_j])}{\sum_i q_i \sum_j (1 - q_j) - \sum_i q_i(1 - q_i)},$$

computed in $O(n \log n)$ by grouping equal scores and cumulating $1 - q$ over sorted scores (a ratio-of-expectations approximation, exact as n grows). A unit test checks it against Monte Carlo draws of the labels.

Calendar-time shocks. The development pattern F is assumed stable. A shock that hits rows of every age at once violates it: during COVID forbearance, loans of all ages became 90 days delinquent within months, which the method reads as the early edge of a much larger wave and projects onto new loans. Actuarial chain-ladder methods share this weakness with “calendar-year effects”.

Hazard variant. We recast the method as an age–period–cohort model in discrete time. On a monthly age grid, row i ’s cumulative incidence without shocks is $p_i F(k)$, which defines monthly hazard increments $L_i(k) = \log(1 - p_i F(k - 1)) - \log(1 - p_i F(k))$. *Age* is fixed by F . *Cohort:* $p_i = \sigma(\text{logit } c_i + \delta_{g(i)})$ with one shift for the newest 12 months of rows and one for the 12 before. *Period:* in each of the last 24 calendar months t , every row alive gets the same extra hazard $\lambda_t \geq 0$. With $S_i(k) = \exp(-\sum_{j \leq k} [L_i(j) + \lambda_{t_i(j)}])$, a positive arriving in month k^* contributes $S_i(k^* - 1) - S_i(k^*)$, a known negative $S_i(H)$, and a pending row $S_i(a_i) - S_i(H) G(a_i)$. Older rows anchor λ : a calendar burst raises their arrivals too, a new-cohort shift does not. This is what separates the two. Two design choices matter. First, $\lambda_t \geq 0$: recent rows have no resolved negatives yet, so with signed period effects “nearly every row will be positive, just slowly” fits as well as the truth. Second, shocks are additive rather than proportional to each row’s hazard, since forbearance-type shocks hit safe and risky rows alike. New rows are projected with the newest cohort’s shift only; bursts are counted but not extrapolated. We fit by L-BFGS-B with analytic gradients (Algorithm 3); Appendix D records the design choices that the model needs to work, several of which we learned by failure.

Algorithm 3 Hazard variant (DA-CBPE-HZ) at decision time s

- 1: $F, G \leftarrow$ arrival curves as in Algorithm 2; $H \leftarrow \lceil q_{0.995}(\text{positive delays}) \rceil$ months
 - 2: $F \leftarrow 0.99 F + 0.01 k/H$ on the monthly grid $k = 0..H$ $\triangleright$ no month with zero arrival mass
 - 3: $W \leftarrow$ informative rows of the last 24 months, minus pending rows past H ; $g(i) \in \{\text{new}, \text{old}\}$ by cohort age < 12 months
 - 4: for each row: calendar month $t_i(k)$ of every age month $k \leq$ current age a_i (the current partial month counted pro rata)
 - 5: $\theta = (\delta_{\text{new}}, \delta_{\text{old}}, \lambda_1.. \lambda_{24})$; $p_i = \sigma(\text{logit } c_i + \delta_{g(i)})$
 - 6: $S_i(k) = \exp(-\sum_{j \leq k} [\log \frac{1-p_i F(j-1)}{1-p_i F(j)} + \lambda_{t_i(j)}])$
 - 7: $\hat{\theta} \leftarrow \arg \min_{\theta} -\sum_{i \in W} \log \ell_i(\theta) + \sum_t \lambda_t^2$ s.t. $\lambda_t \geq 0$ $\triangleright$ L-BFGS-B, analytic gradient, warm start
 $\ell_i = S_i(k_i^* - 1) - S_i(\min(k_i^*, a_i))$ (positive), $S_i(H)$ (negative), $S_i(a_i) - S_i(H) G(a_i)$ (pending)
 - 8: **new batch:** CBPE with $p_i(\delta_{\text{new}})$ $\triangleright$ bursts not projected forward
 - 9: **in-flight book:** arrived labels as observed; pending rows $q_i = \frac{S_i(a_i) - S_i(H)}{S_i(a_i) - S_i(H) G(a_i)}$ with fitted λ
-

Table 2: Degradation detection: medians over 16 cases.

Detector	Spearman ρ	Alarm rate	Precision	Recall
Delay-adjusted CBPE alarm (ours)	0.58	0.32	0.84	0.52
CBPE alarm	0.50	0.11	0.82	0.11
ADWIN on scores	0.50	0.10	0.67	0.13
KS on scores	0.49	1.00	0.48	1.00
PSI	0.40	1.00	0.44	1.00
MMD	0.38	1.00	0.52	1.00
Univariate KS / χ^2	0.37	1.00	0.48	1.00
Domain classifier	0.37	1.00	0.52	1.00
Latest complete cohort alarm	0.31	0.39	0.71	0.71
Recent arrivals alarm	0.21	0.24	0.63	0.40
ADWIN on arrived errors	0.17	0.13	0.83	0.19

6 Results

Detection (T3). Table 2 summarizes 16 cases (8 experiments $\times$ 2 event types). Every input-drift test alarms in every period of the median case: lender churn, interest-rate levels, a discontinued channel code and a field introduced in 2017 all shift Freddie Mac’s inputs without necessarily hurting the model. Their scores still rank severity moderately (median ρ 0.37–0.49), but their thresholds are unusable. The delay-adjusted alarm both ranks degradation best (median ρ 0.58) and alarms selectively (32% of periods, precision 0.84). ADWIN on arriving errors, the classic streaming setup, ranks last: under delay it sees too little, too late. Figure 1 shows the trade-off; per-experiment correlations and alarm statistics are in Appendix E (Tables 7 and 8).

Estimation (T1, T2). Tables 3 and 4 report MAE over five seeds. Seed-to-seed variation is small relative to the gaps between methods.

Next batch. No method dominates. Label-free and delay-adjusted estimators are most accurate when shift is mostly in the inputs (TabReD homesite and homecredit, BRFS). With short delays and a drifting label, simply using the latest complete labels wins (ACS income under wage inflation, NYC taxi tips, TabReD ecom). On short-horizon streams the hazard variant’s extra parameters are poorly anchored and cost accuracy (NYC taxi positive rate: 0.030 vs. 0.012 for the chain-ladder

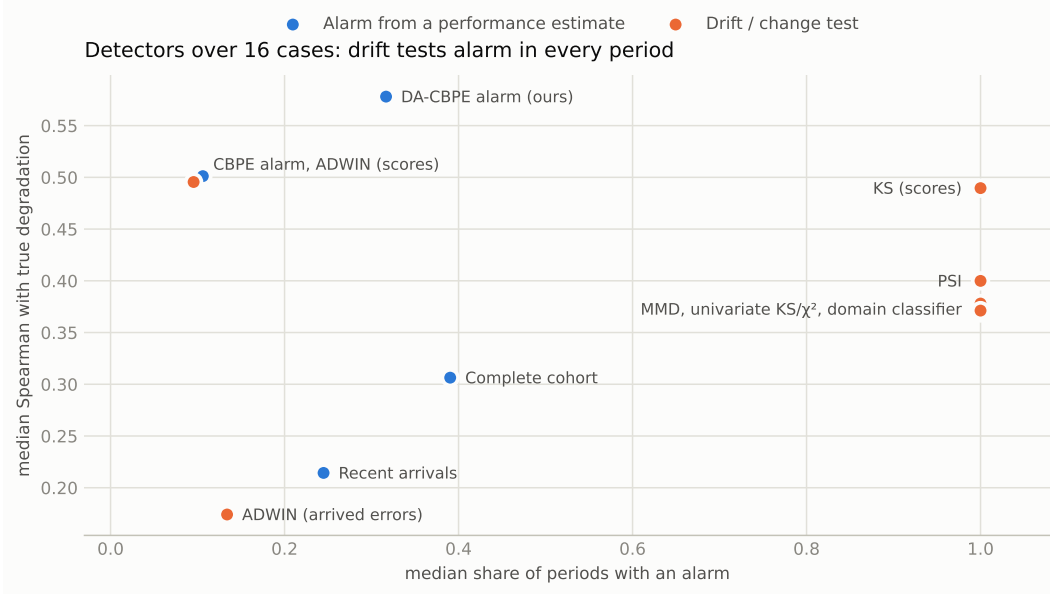


Figure 1: Each detector’s median alarm rate against its median rank correlation with true degradation, over 16 cases. Every input-drift test sits at an alarm rate of 1.0.

form), so it should be preferred only where labels arrive over a long horizon. On Freddie Mac the task is close to unanswerable: loans made in 2019Q1 defaulted at 4.4% because of a pandemic a year later.

In-flight book. Here early labels pay off (Table 4). The hazard variant has the lowest error in three of four cases. On *freddie_2007* its default-rate error is 0.0060 against 0.0087 for the best label-based baseline, and its AUC error 0.0180 against 0.0215; block-bootstrap intervals of the paired differences exclude zero (default rate: -0.0034 $[-0.0076, -0.0004]$ vs. waiting for complete cohorts; AUC: -0.0058 $[-0.0126, -0.0001]$ vs. labels to date). The exception is *freddie_2016* AUC, where labels to date is better (0.0263 vs. 0.0287; interval includes zero). All paired comparisons are in Table 10; the accuracy-only estimators ATC and DoC are compared in Table 11.

Figure 2 shows the mechanism. CBPE stays near the reference default rate through the 2007–08 crisis (loans made in 2008Q1: true 6.6%, CBPE 1.0%). Waiting for complete cohorts sees the crisis about two years late and then reports it after it has passed (4.7% for 2010Q1 loans against a true 0.6%). On the in-flight book both delay-adjusted estimators react within quarters. Under COVID, the chain-ladder form reads forbearance as the start of a much larger wave (peak in-flight estimate 8.3% against a true peak of 4.3%); the hazard variant attributes it to calendar months and peaks at 4.3%.

Synthetic checks. Two controlled streams isolate the two kinds of shock: a lasting shift (the outcome rate of new cohorts jumps) and a calendar burst (for four months every active loan has an extra 3% monthly default hazard, new cohorts unaffected). The chain-ladder form tracks the lasting shift best (in-flight MAE 0.007 vs. 0.050 for complete cohorts) but its error grows more than tenfold under the burst (0.094). The hazard variant gives up a little on the lasting shift (0.010) and cuts the burst error to 0.030, the best of all methods.

Table 3: Next-batch estimation. MAE, mean (std) over 5 seeds; lower is better, best in bold.

Experiment	Metric	Ref.	Recent	Complete	CBPE	DA-CBPE	DA-CBPE-HZ
acs_income_2016	accuracy	0.0369 (0.0018)	–	0.0087 (0.0002)	0.0314 (0.0024)	0.0145 (0.0020)	0.0154 (0.0021)
acs_income_2016	pos. rate	0.0929 (0.0015)	–	0.0199 (0.0002)	0.0778 (0.0022)	0.0178 (0.0006)	0.0202 (0.0014)
acs_income_2016	AUC	0.0096 (0.0020)	–	0.0019 (0.0003)	0.0071 (0.0021)	0.0091 (0.0022)	0.0090 (0.0022)
brfss_2014	pos. rate	0.0093 (0.0009)	0.0045 (0.0002)	0.0052 (0.0002)	0.0041 (0.0010)	0.0033 (0.0004)	0.0039 (0.0006)
brfss_2014	AUC	0.0106 (0.0007)	0.0042 (0.0003)	0.0062 (0.0003)	0.0050 (0.0001)	0.0051 (0.0001)	0.0051 (0.0001)
freddie_2007	pos. rate	0.0108 (0.0000)	0.0140 (0.0000)	0.0166 (0.0000)	0.0123 (0.0003)	0.0143 (0.0002)	0.0122 (0.0009)
freddie_2007	AUC	0.0498 (0.0034)	0.0502 (0.0033)	0.0471 (0.0025)	0.0382 (0.0075)	0.0360 (0.0063)	0.0375 (0.0069)
freddie_2016	pos. rate	0.0140 (0.0000)	0.0166 (0.0000)	0.0174 (0.0000)	0.0140 (0.0004)	0.0218 (0.0010)	0.0142 (0.0011)
freddie_2016	AUC	0.1054 (0.0063)	0.0724 (0.0077)	0.0422 (0.0029)	0.0726 (0.0044)	0.0691 (0.0038)	0.0713 (0.0039)
tabred_ecom	pos. rate	0.0697 (0.0019)	0.0772 (0.0000)	0.0966 (0.0000)	0.1741 (0.0094)	0.1449 (0.0075)	0.1717 (0.0213)
tabred_ecom	AUC	0.2316 (0.0255)	0.1014 (0.0128)	0.0717 (0.0154)	0.1133 (0.0288)	0.1089 (0.0294)	0.1083 (0.0292)
tabred_homecredit	pos. rate	0.0094 (0.0000)	0.0119 (0.0000)	0.0099 (0.0000)	0.0089 (0.0019)	0.0089 (0.0006)	0.0086 (0.0005)
tabred_homecredit	AUC	0.0180 (0.0027)	0.0254 (0.0026)	0.0185 (0.0034)	0.0159 (0.0018)	0.0159 (0.0017)	0.0159 (0.0017)
tabred_homesite	pos. rate	0.0236 (0.0016)	0.0160 (0.0000)	0.0160 (0.0000)	0.0052 (0.0013)	0.0046 (0.0015)	0.0051 (0.0017)
tabred_homesite	AUC	0.0137 (0.0017)	0.0058 (0.0001)	0.0056 (0.0003)	0.0051 (0.0020)	0.0048 (0.0020)	0.0049 (0.0021)
tlc_2019	accuracy	0.0062 (0.0005)	0.0036 (0.0001)	0.0031 (0.0000)	0.0055 (0.0004)	0.0086 (0.0003)	0.0144 (0.0005)
tlc_2019	pos. rate	0.0339 (0.0013)	0.0085 (0.0000)	0.0078 (0.0000)	0.0164 (0.0010)	0.0116 (0.0002)	0.0301 (0.0007)
tlc_2019	AUC	0.0189 (0.0009)	0.0060 (0.0002)	0.0052 (0.0001)	0.0057 (0.0002)	0.0078 (0.0003)	0.0073 (0.0001)

Table 4: In-flight book estimation (Freddie Mac, natural label delay). MAE, mean (std) over 5 seeds; lower is better, best in bold.

Experiment	Metric	Ref.	To date	Complete	CBPE	DA-CBPE	DA-CBPE-HZ
freddie_2007	pos. rate	0.0102 (0.0000)	0.0090 (0.0000)	0.0087 (0.0000)	0.0119 (0.0003)	0.0065 (0.0000)	0.0060 (0.0008)
freddie_2007	AUC	0.0462 (0.0035)	0.0215 (0.0009)	0.0394 (0.0016)	0.0333 (0.0086)	0.0220 (0.0046)	0.0180 (0.0021)
freddie_2016	pos. rate	0.0124 (0.0000)	0.0088 (0.0000)	0.0083 (0.0000)	0.0127 (0.0002)	0.0103 (0.0001)	0.0060 (0.0003)
freddie_2016	AUC	0.0968 (0.0058)	0.0263 (0.0012)	0.0355 (0.0018)	0.0692 (0.0037)	0.0422 (0.0023)	0.0287 (0.0024)

Retraining (T4). At moderate retraining cost ($\rho = 1$) a fixed calendar is the best policy in 5 of 8 experiments (with the period chosen in hindsight, which flatters it). Monitoring helps on Freddie Mac: retraining on alarms from recently arrived labels wins on freddie_2007, and on CBPE alarms on freddie_2016. Retraining on input-drift alarms is equivalent to always retraining, since those tests always fire: cheapest when retraining is free, among the worst once it costs anything. On tabred_ecom retraining never pays: newer models train on too little data. Full regret tables are in Appendix E (Table 9).

7 Software

The benchmark is a Python package, `lfmm`, released under the MIT license with all experiment scripts.¹ It has four parts: downloaders that record every source file’s URL and SHA-256 checksum; stream loaders that build the label and its arrival time from the raw files; the four tasks; and the methods, each a small class with `fit` on the reference window and one method that answers the task. Listing 1 runs the estimation task on any stream; adding a method means subclassing `Estimator`, `Detector` or `Policy` (Appendix A). Every listing in this paper is a runnable file in the repository’s `paper/snippets/` folder.

```
from lfmm.estimators import CBPE, DelayAdjustedCBPE, LatestCompleteCohort
from lfmm.harness import run_estimation

# Train on fully labeled cohorts before 2023, freeze, walk forward monthly.
result = run_estimation(
    stream, train_end="2023-01-01", freq="M", metrics=("roc_auc", "prevalence"),
```

¹<https://github.com/rajkumar160798/label-free-model-monitoring>

Freddie Mac, model built Jan 2007: estimated vs true default rate

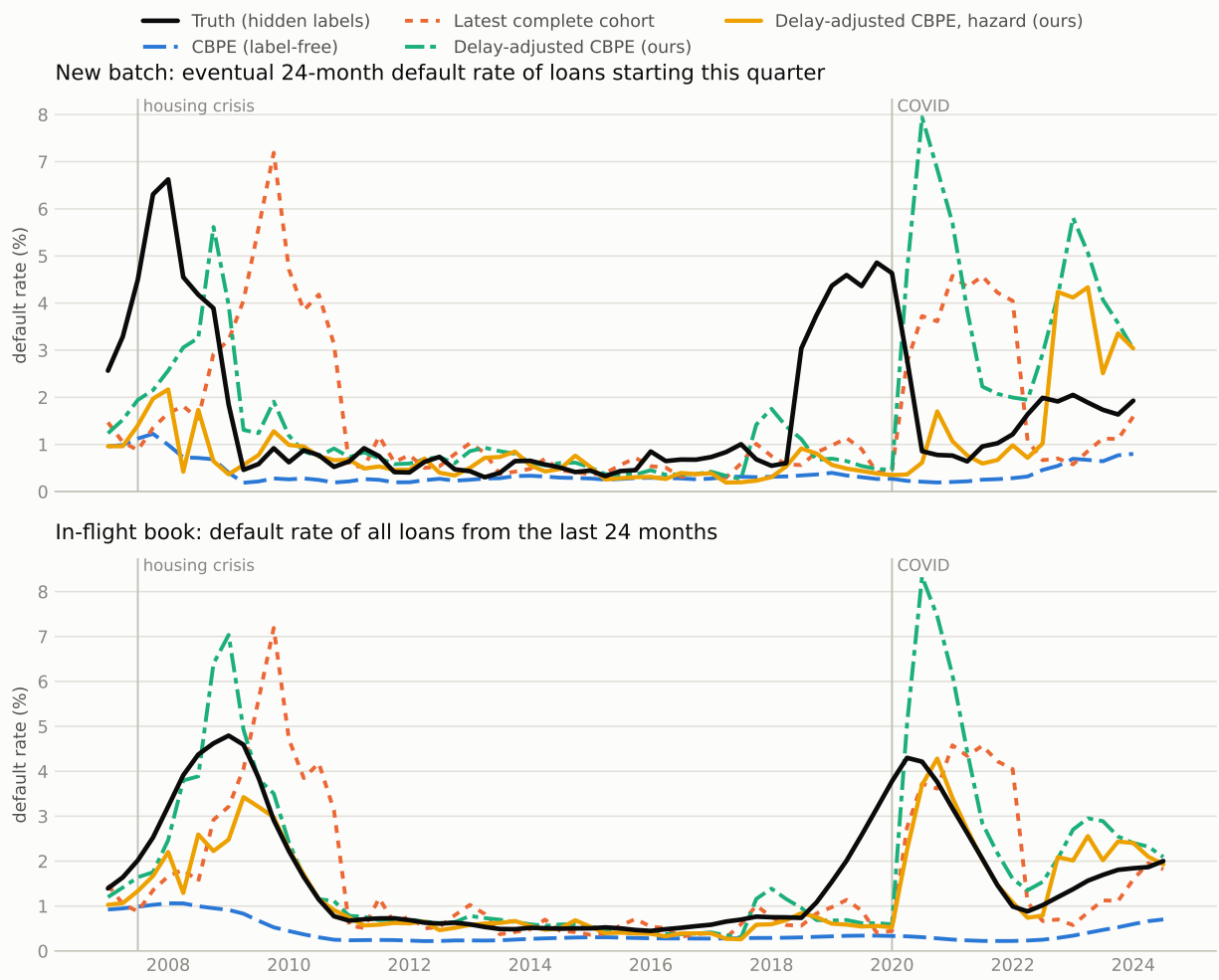


Figure 2: Freddie Mac, model built January 2007. Top: eventual default rate of each quarter’s new loans. Bottom: default rate of the in-flight book (loans from the last 24 months).

```

    estimators=[CBPE(), DelayAdjustedCBPE(), LatestCompleteCohort(freq="M")],
)
print(result.summary())                # MAE and bias per estimator
print(result.bootstrap_mae("prevalence")) # 95% block-bootstrap intervals
print(result.compare("prevalence", "da_cbpe", "cbpe"))

```

Listing 1: Estimating performance on a stream with delayed labels.

Everything runs on a 16-core desktop without a GPU. Single task runs take from under a minute to under half an hour per experiment (Appendix C); the hazard estimator and the domain-classifier and MMD detectors dominate the time.

8 Limitations

Simulated delays on six of eight experiments are constant; only Freddie Mac has natural arrival. Estimation results cover five seeds (model randomness and, for sampled streams, the sample); detection and retraining results are single runs with period-level summaries. The fixed-calendar policy’s period is chosen in hindsight. Regression tasks (five TabReD datasets) are not yet supported. Freddie Mac and TabReD data cannot be redistributed; the benchmark ships loaders and checksums, not data.

9 Conclusion

Monitoring benchmarks should model when labels arrive. Doing so shows that input-drift alarms are uninformative on real data, that label-free and label-based estimators fail in opposite directions, and that early-arriving labels, once their bias is corrected, close much of the gap.

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A Using the benchmark

Every listing below is a file in `paper/snippets/` and runs as shown against the released package (it uses a small synthetic stream so it runs in seconds without downloading data).

```
import numpy as np
import pandas as pd
from lfmm.streams.base import Stream, with_label_delay

# Any table of scored events becomes a stream: features, label,
# when each row is scored, and when its label becomes known.
rng = np.random.default_rng(0)
n = 24_000
month = np.repeat(np.arange(24), n // 24)
X = pd.DataFrame({"x1": rng.normal(month / 12, 1), "x2": rng.normal(0, 1, n)})
y = (rng.random(n) < 1 / (1 + np.exp(-(X.x1 - X.x2 - 1))))).astype(float)
t = (np.datetime64("2022-01", "M") + month).astype("datetime64[ns]")

stream = Stream("demo", X, y, event_time=t, label_time=t)
stream = with_label_delay(stream, np.timedelta64(60, "D")) # labels 60 days late
print(stream.summary())
```

Listing 2: Building a stream from any table of scored events (`s1_stream.py`).

```
import numpy as np
from s1_stream import stream
from lfmm.estimators import Estimator, History
from lfmm.harness import run_estimation

class RecentPositiveRate(Estimator):
    """Toy method: positive rate among labels that arrived in the last 90 days."""
    name = "recent_rate"
    supports = ("prevalence",)

    def estimate(self, metric: str, proba: np.ndarray, history: History) -> float:
        arrived = ~np.isnat(history.label_time)
        recent = arrived & (history.label_time > history.now - np.timedelta64(90, "D"))
        return float(np.nanmean(history.y_known[recent])) if recent.any() else np.nan

res = run_estimation(stream, "2023-01-01", freq="M", metrics=("prevalence",),
                    estimators=[RecentPositiveRate()])
print(res.summary())
```

Listing 3: Adding an estimator: implement `estimate`, which sees the batch's scores and a `History` holding only labels that have arrived (`s3_custom.py`).

```
from s1_stream import stream
from lfmm.detectors import PSI, EstimatorAlarm
from lfmm.estimators import DelayAdjustedCBPE
from lfmm.harness import run_detection_events
from lfmm.retrain import Always, Every, Never, OnAlarm, build_bank, evaluate_policies

# Detection: score each detector against "AUC dropped by more than delta".
events = {"roc_auc": ((0.01, 0.02), 0.02)}
detectors = [PSI(), EstimatorAlarm(DelayAdjustedCBPE(), "roc_auc", 0.02)]
results = run_detection_events(stream, "2023-01-01", detectors, events, freq="M")
print(results["roc_auc"].summary()[["spearman", "precision", "recall", "alarm_rate"]])
```

```
# Retrain or not: one candidate model per month, then any policy by lookup.
bank = build_bank(stream, "2023-01-01", "M", verbose=False)
regret = evaluate_policies(bank, [Never(), Always(), Every(3), OnAlarm(PSI)])
print(regret[regret.rho == 1.0][["policy", "retrains", "regret"]])
```

Listing 4: Detection against an event definition, and the retrain-or-not task with fixed and alarm-triggered policies (s4_detect_retrain.py).

Running the paper’s experiments. The eight experiments are declared in `lfmm.experiments.EXPERIMENTS` (stream loader, training date, period length, metrics and event definitions). Each task has a script: `run_estimation.py`, `run_inflight.py`, `run_detection.py`, `run_retrain.py` and `run_seeds.py`; tables and figures are regenerated from their CSV outputs by `make_tables.py`, `make_appendix.py` and `make_figures.py`. Raw data are fetched by `python -m lfmm.download`, which writes a manifest with the source URL and SHA-256 of every file; Freddie Mac and TabReD files, which need an account or accepted terms, are downloaded by hand and registered with the same command.

B Dataset construction

Table 5: Streams after construction. Rows are those used in the experiments (ACS and BRFSS are sampled per year from the full tables; NYC taxi keeps 50k card trips per month).

Stream	Source	Years	Rows used	Num.	Cat.	Target	Positive rate
Freddie Mac	SFLLD sample	1999–2026	1,362,496	10	12	default within 24 months	0.4%–4.8% by year
ACS income	Census PUMS	2014–2024	1.0M (of 16.6M)	5	6	income > \$50k	31.7%–49.1%
BRFSS	CDC survey	2011–2024	1.68M (of 6.3M)	8	17	diagnosed diabetes	12.4%–14.6%
NYC taxi	TLC trip records	2019–2021	1.8M (of 140M)	13	4	tip ≥ 25% of fare	54.8%–61.9%
TabReD homecredit	TabReD	2019–2020	381,664	612	82+2 bin.	loan default	3.2%
TabReD homesite	TabReD	2013–2015	260,753	253	23+23 bin.	quote conversion	18.8%
TabReD ecom	TabReD	2013	160,057	113	6 bin.	repeat purchase	27.1%

Freddie Mac. Origination features: credit score, mortgage-insurance percentage, number of units, combined and plain loan-to-value, debt-to-income, original balance, interest rate, term and number of borrowers (numeric); first-time buyer, occupancy, channel, state, property type, loan purpose, seller, super-conforming, program, relief refinance, valuation method and interest-only flags (categorical). Sentinel codes (e.g. 9999 for credit score, 999 for ratios) are set to missing. Identifiers, maturity date, MSA and postal code are dropped, and so is MI cancellation, which is observed after origination and would leak. The number-of-borrowers field changed coding in 2018Q2 (“2 or more” became exact counts) and is capped at 2. The event is delinquency of 90 days or more, REO acquisition, or a zero-balance code for third-party sale, short sale, REO disposition or whole-loan sale, within 24 calendar months of the first payment. A negative is known at month 24 or at an earlier prepayment (zero-balance code 01); loans with neither never resolve. Four seller-owned modified loans whose first-payment date was reset years after origination are dropped. Counting the window by the reported loan age instead of calendar months would be wrong: Freddie Mac resets loan age at modification, which in a first version of this benchmark turned 17.7% of positives into defaults that happened years later.

ACS. The Folktables ACSIncome task [4]: employed adults (age > 16, income > \$100, hours > 0, weight ≥ 1), features age, education, hours, occupation, place of birth (numeric) and class of

worker, marital status, relationship, sex, race and state (categorical). From 2019 the relationship field RELP was replaced by RELSHIPP with new codes; we map RELSHIPP to RELP. The state column is ST until 2022 and STATE from 2023. Occupation codes switched to the 2018 SOC scheme in 2018 and are left as they are, as a production pipeline would see them. Income is nominal, so the positive rate rises with wages.

BRFSS. Interviews are ordered by interview month (a survey year’s last interviews fall in the next calendar year). Variables renamed across years are merged (diabetes, sex, income, kidney disease, depression, arthritis, cost barrier, personal doctor, race). From 2021 the income variable splits the top band (\$50k+) into three; we cap it back to five bands. Employment is not asked in 2011–2012 and health insurance is not asked of all ages in 2021–2022; those values stay missing. Positive: ever told diabetes (excluding pregnancy-only); “don’t know” and refusals are dropped.

NYC taxi. Credit-card trips only (cash tips are not recorded) with positive fare and distance, duration 1–180 minutes, and pickup and drop-off within the file’s month. Features: passengers, distance, fare, surcharges and tolls, duration, hour, weekday, pickup and drop-off zone, vendor, rate code and store-and-forward flag. The total amount is excluded because it includes the tip; the airport fee exists only from 2021. The positive rate jumps from 25% in January 2019 to 58% from February, coinciding with the congestion surcharge introduced on 1 February 2019 (we have not confirmed the cause), so the training window starts in February.

TabReD. The authors’ preprocessed arrays [18]; rows ordered by the timestamp in the first metadata column, which is in microseconds for homesite and in days for homecredit and ecom. TabReD’s own splits are not used: the walk-forward protocol replaces them.

C Implementation details

Model. Scikit-learn’s histogram gradient boosting (300 iterations, learning rate 0.08, early stopping on 10% of the training rows, native categorical handling; categoricals with more than 250 levels become integer codes). Features that are missing or constant in the training rows are dropped. At most 500k training rows (300k for the retraining bank).

Protocol. Trainable cohorts need $\geq 95\%$ of labels arrived; reference window = the last 12 months of trainable cohorts (a random 30% when that would leave too little training data); truth requires $\geq 90\%$ of a batch’s labels to resolve. Retraining uses the last 60 months of trainable cohorts.

Estimators. Recent arrivals: labels arrived in the last 90 days (≥ 200 rows). Latest complete cohort: newest monthly cohort with $\geq 95\%$ of labels arrived. CBPE: isotonic calibration on the reference window. ATC and DoC: as published, with confidence $\max(p, 1 - p)$. DA-CBPE: 12-month fitting window, 36 months of mature rows for the arrival curves, up to 50k rows. Hazard variant: 24-month window, cohort split at 12 months, 24 calendar months, ridge 1, up to 20k rows, $\lambda_t \in [0, 0.5]$, $\delta \in [-5, 5]$.

Detectors. Univariate: KS (numeric) and χ^2 (categorical) with Bonferroni correction, scored by the largest effect size. PSI: 10 reference-quantile bins plus a missing bin, alarm at 0.25. KS on scores: alarm at $p < 0.05$. Domain classifier: gradient boosting (100 iterations) on up to 5k reference and 5k batch rows, 3-fold cross-validated AUC, alarm at 0.6. MMD: RBF kernel

with median bandwidth on standardized numeric features and the score, 1k rows per side, 100 permutations, alarm at $p < 0.05$. ADWIN: $\delta = 0.002$, up to 2k values per batch. Estimator alarms fire when the estimate is worse than the reference by more than the experiment’s alarm Δ : AUC 0.05 and default rate 0.01 for Freddie Mac; accuracy 0.03 and positive rate 0.05 for ACS; AUC 0.02 and positive rate 0.01 for TabReD homecredit; AUC 0.02 and positive rate 0.02 for the others.

Statistics. Moving-block bootstrap over periods, block length 4, 2000 resamples; paired comparisons resample the per-period difference. Estimation is repeated over 5 seeds (model randomness, training subsample, and for ACS and BRFS the per-year sample).

Compute. One 16-core Windows desktop, no GPU. Measured times: next-batch estimation under a minute per experiment without the hazard estimator; one hazard fit 3–11 seconds per Freddie Mac quarter; detection on both Freddie Mac experiments about 20 minutes; the retraining bank for freddie_2016 about 8 minutes (33 models).

D Designing the hazard variant

The hazard variant took several attempts; each failure is instructive for anyone extending it. Table 6 lists every version we tried on the two synthetic streams of Section 6, in the order we tried them.

Table 6: Design ablation on synthetic data (MAE of the positive rate). Each row adds one change to the previous one. The burst stream needs the calendar effect; the lasting-shift stream checks that fixing it does not break what already worked.

Version	Calendar burst		Lasting shift	
	next batch	in-flight	next batch	in-flight
Chain-ladder DA-CBPE	0.133	0.094	0.048	0.007
+ arrival-speed effect κ , last 3 months	0.159	0.127	0.053	0.011
+ κ_t per calendar month (ridge)	0.187	0.151	0.060	0.020
Hazard model, proportional shocks, signed κ_t	0.181	0.136	0.059	0.014
+ shocks constrained to $\kappa_t \geq 0$	0.080	0.056	0.052	0.009
+ additive shocks, two cohort shifts, 24-month window (final)	0.039	0.030	0.061	0.010
Latest complete cohort (baseline)	0.082	0.053	0.110	0.050
CBPE (baseline)	0.030	0.055	0.125	0.065

1. **A shock must be able to add positives, not only make them arrive sooner.** The first two versions only changed arrival timing; whether a row is positive could then only come from δ , so the burst was still read as a lasting shift.
2. **Shocks must be non-negative.** Recent rows have no resolved negatives yet. With signed period effects, “nearly every row will be positive, just slowly” explains the data as well as the truth, and the optimizer drove δ to its bound.
3. **Shocks should be additive.** A forbearance-type shock adds a similar hazard to safe and risky loans; a proportional shock cannot reach safe loans and pushes the excess back into δ .
4. **Older cohorts identify the shock.** A calendar burst raises arrivals for all ages, a cohort shift only for new loans. With a 24-month window and separate shifts for new and older cohorts, the two are separable; with 12 months of young loans they are not.

5. **Survival must be credited to the exact current age.** Flooring ages to whole months counted arrivals in the current month without crediting the survival that preceded them; that alone biased δ up by about 0.15 (with shocks switched off, the model should and now does reproduce the chain-ladder δ to within 0.01).
6. **Real arrival curves need care.** On Freddie Mac, the empirical curve has months with no arrival mass (defaults almost never arrive in month 1), which gave some arrivals probability zero and broke the line search; we blend in 1% of a uniform curve. The grid length uses the 99.5th percentile of delays, and pending rows that should already have resolved are left out.

E Additional results

Table 7: Degradation detection, per experiment: Spearman correlation between the detector’s drift score and the true degradation (higher is better). *perf*: AUC drop (accuracy for ACS); *rate*: change in positive rate. – = score constant or undefined.

Detector	FM07		FM16		ACS		BRFSS		Taxi		HCred		HSite		Ecom	
	perf	rate	perf	rate	perf	rate	perf	rate	perf	rate	perf	rate	perf	rate	perf	rate
DA-CBPE alarm (ours)	0.57	0.23	0.33	0.37	1.00	1.00	0.75	0.74	0.80	0.74	0.73	0.69	0.67	0.83	0.60	0.10
CBPE alarm	0.57	-0.15	0.21	0.39	0.93	0.93	0.75	0.78	0.90	0.86	0.72	0.57	0.66	0.83	-0.20	-0.50
Complete-cohort alarm	0.55	0.09	0.27	0.09	1.00	1.00	0.54	0.35	0.86	0.88	0.00	0.04	0.79	0.45	–	–
Recent-arrivals alarm	0.29	0.02	0.23	-0.02	–	–	0.78	0.42	0.85	0.89	0.06	0.13	0.68	0.25	0.20	0.50
ADWIN (scores)	-0.42	-0.54	-0.04	-0.10	0.98	0.98	0.18	0.59	0.81	0.79	0.65	0.81	0.49	0.50	-0.20	-0.50
ADWIN (arrived errors)	0.05	-0.04	0.16	-0.16	0.90	0.90	0.82	0.47	0.54	0.50	-0.30	-0.39	0.52	0.19	-0.60	-0.10
KS (scores)	-0.34	-0.58	0.29	0.34	0.93	0.93	0.54	0.82	0.91	0.90	0.44	0.55	0.85	0.41	-0.20	-0.50
Univariate KS/ χ^2	0.21	-0.13	0.39	0.35	1.00	1.00	0.42	0.27	0.73	0.65	–	–	-0.10	0.34	–	–
PSI (max)	0.36	0.10	0.43	0.39	1.00	1.00	0.41	0.21	0.17	0.44	0.58	0.44	0.41	0.32	-0.22	-0.45
MMD	0.15	-0.35	0.37	0.39	0.83	0.83	0.85	0.50	0.90	0.76	-0.60	-0.17	0.85	0.09	0.10	0.10
Domain classifier	0.15	-0.25	0.40	0.39	0.95	0.95	0.66	0.35	0.18	0.40	-0.17	-0.17	–	–	–	–

Table 8: Alarm behaviour of each detector over the 16 detection cases.

Detector	Median alarm rate	Cases alarming every period	Median precision	Median recall
DA-CBPE alarm (ours)	0.32	0%	0.84	0.52
CBPE alarm	0.11	6%	0.82	0.11
Complete-cohort alarm	0.39	12%	0.71	0.71
Recent-arrivals alarm	0.24	0%	0.63	0.40
ADWIN (scores)	0.10	12%	0.67	0.13
ADWIN (arrived errors)	0.13	12%	0.83	0.19
KS (scores)	1.00	75%	0.48	1.00
Univariate KS/ χ^2	1.00	100%	0.48	1.00
PSI (max)	1.00	75%	0.44	1.00
MMD	1.00	62%	0.52	1.00
Domain classifier	1.00	75%	0.52	1.00

F Data access and licenses

ACS, BRFSS and NYC taxi records are US public-sector releases. Freddie Mac’s loan-level data are free for non-commercial research but may not be redistributed; TabReD combines Yandex data for non-commercial research with Kaggle competition data under their rules. The repository therefore contains no raw data: only download code, checksums, loaders and aggregated results. All sources are de-identified public releases; the benchmark adds no personal data.

Table 9: Retrain or not: regret against the hindsight-optimal schedule at retrain cost $\rho = 1$ (number of retrains in parentheses). For *every* k , the best k is chosen in hindsight. Lower is better; best non-oracle policy in bold.

Policy	FM07	FM16	ACS	BRFSS	Taxi	HCred	HSite	Ecom
Hindsight oracle	0.000 (5)	0.000 (5)	0.000 (2)	0.000 (5)	0.000 (3)	0.000 (3)	0.000 (3)	0.000 (0)
Never	0.745 (0)	0.174 (0)	0.107 (0)	0.136 (0)	0.029 (0)	0.070 (0)	0.132 (0)	0.000 (0)
Always	0.745 (68)	0.174 (32)	0.107 (7)	0.136 (44)	0.029 (29)	0.070 (13)	0.132 (19)	0.168 (4)
Every k (best k)	0.432 (8)	0.136 (16)	0.011 (2)	0.011 (5)	0.012 (2)	0.018 (2)	0.022 (3)	0.087 (1)
Alarm: complete cohort	0.488 (11)	0.183 (7)	0.024 (1)	0.049 (2)	0.021 (13)	0.070 (0)	0.082 (4)	0.068 (1)
Alarm: recent arrivals	0.423 (16)	0.109 (9)	0.107 (0)	0.070 (1)	0.021 (12)	0.070 (0)	0.110 (4)	0.156 (2)
Alarm: CBPE	0.745 (0)	0.091 (1)	0.107 (0)	0.136 (0)	0.020 (9)	0.070 (0)	0.068 (4)	0.162 (3)
Alarm: PSI	0.745 (68)	0.174 (32)	0.107 (0)	0.083 (27)	0.017 (14)	0.070 (13)	0.132 (19)	0.168 (4)
Alarm: KS (scores)	0.745 (68)	0.174 (32)	0.066 (5)	0.122 (39)	0.029 (29)	0.070 (13)	0.132 (19)	0.168 (4)
Alarm: domain classifier	0.745 (68)	0.174 (32)	0.026 (2)	0.055 (20)	0.024 (20)	0.070 (13)	0.132 (19)	0.168 (4)

Table 10: In-flight book: MAE of the hazard variant minus MAE of each baseline (negative = ours better), with 95% moving-block bootstrap intervals over quarters (block length 4, 2000 resamples). Bold: interval excludes zero.

Experiment	Metric	vs. complete cohort	vs. labels to date	vs. CBPE	vs. DA-CBPE
freddie_2007	default rate	-0.0034 [-0.0076, -0.0004]	-0.0038 [-0.0049, -0.0025]	-0.0070 [-0.0105, -0.0041]	-0.0013 [-0.0045, +0.0013]
freddie_2007	AUC	-0.0220 [-0.0322, -0.0137]	-0.0058 [-0.0126, -0.0001]	-0.0130 [-0.0252, -0.0033]	-0.0031 [-0.0072, +0.0001]
freddie_2016	default rate	-0.0025 [-0.0082, +0.0012]	-0.0029 [-0.0036, -0.0019]	-0.0069 [-0.0130, -0.0025]	-0.0045 [-0.0103, -0.0009]
freddie_2016	AUC	-0.0032 [-0.0164, +0.0104]	+0.0041 [-0.0056, +0.0114]	-0.0411 [-0.0654, -0.0219]	-0.0134 [-0.0224, -0.0068]

Table 11: Accuracy estimation, including the accuracy-only estimators ATC and DoC (balanced-class streams). MAE, mean (std) over 5 seeds.

Experiment	Ref.	Complete	CBPE	ATC	DoC	DA-CBPE	DA-CBPE-HZ
acs_income_2016	0.0369 (0.0018)	0.0087 (0.0002)	0.0314 (0.0024)	0.0294 (0.0045)	0.0311 (0.0022)	0.0145 (0.0020)	0.0154 (0.0021)
tlc_2019	0.0062 (0.0005)	0.0031 (0.0000)	0.0055 (0.0004)	0.0120 (0.0051)	0.0053 (0.0004)	0.0086 (0.0003)	0.0144 (0.0005)

Freddie Mac, model built Jan 2016: estimated vs true default rate

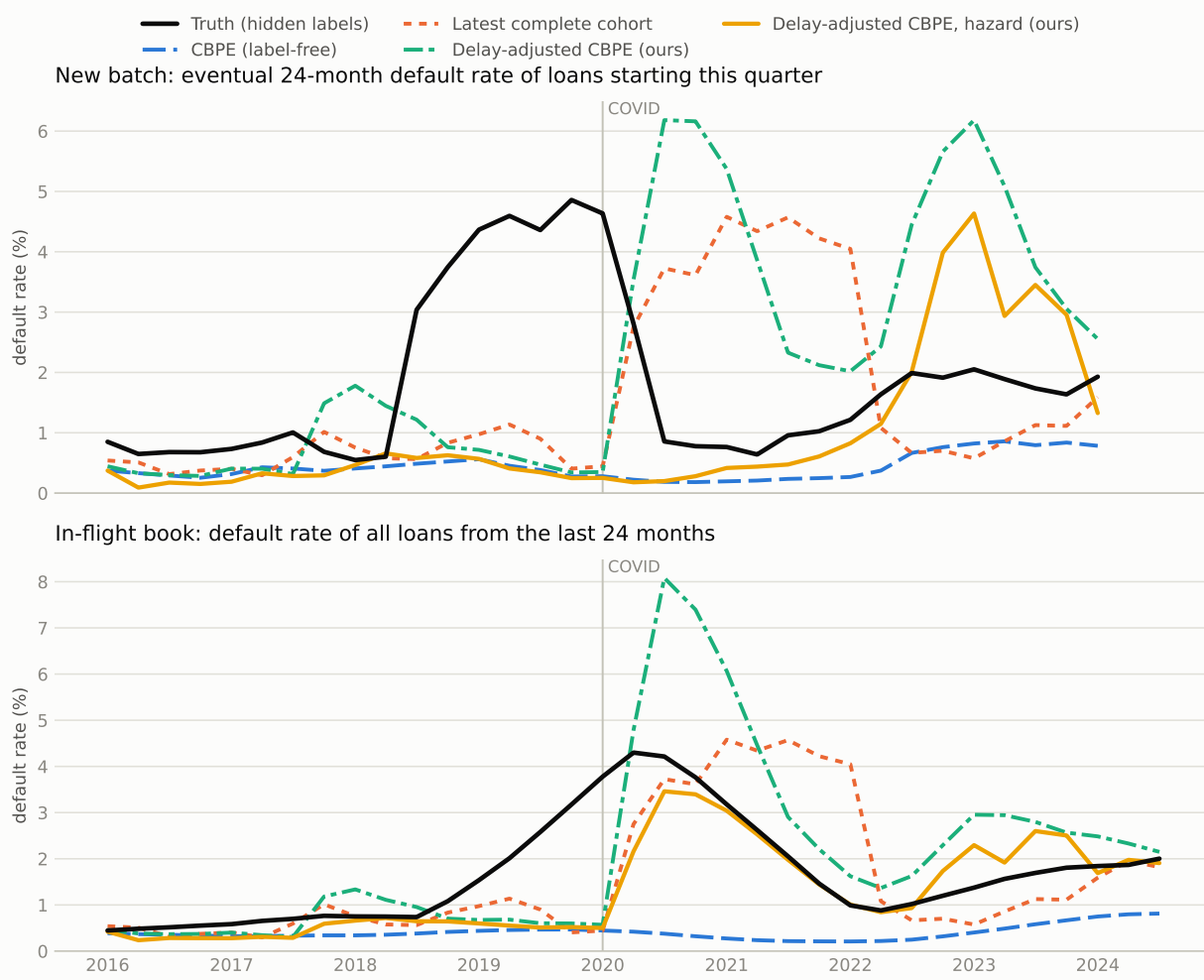


Figure 3: Freddie Mac, model built January 2016 (COVID is the main shock). Top: next-batch default rate; bottom: in-flight book.

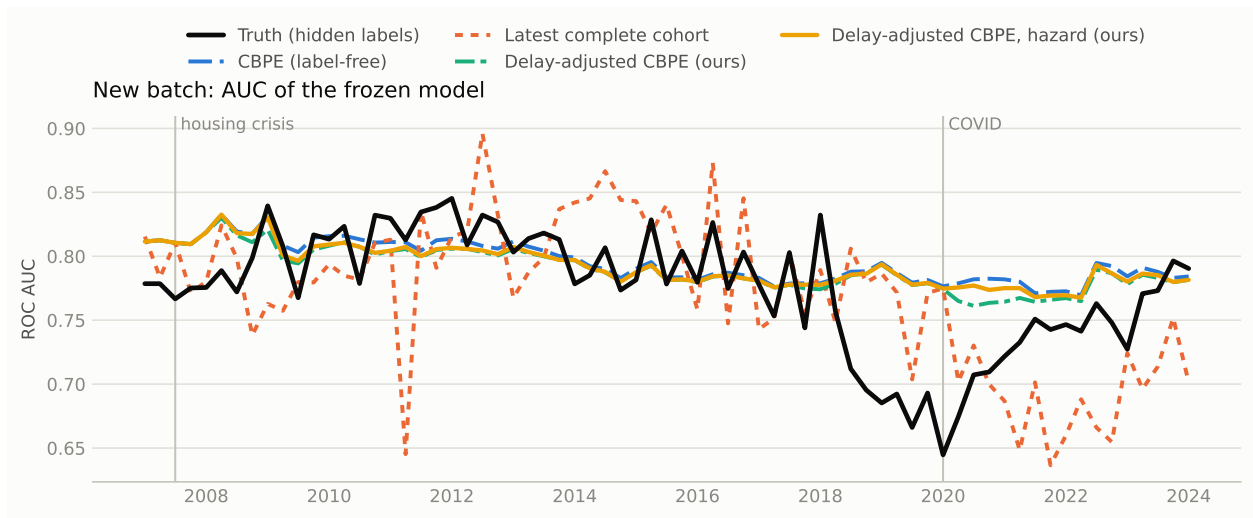


Figure 4: Freddie Mac, model built January 2007: next-batch AUC. AUC moves less than the default rate; label-free estimates of it stay smooth while the complete-cohort baseline is noisy.